

AI Based Green Budget Optimizer

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Abstract—The ever-increasing world environmental problem requires decisive, mass climate action. Nevertheless, massive fiscal limitations and shortcomings of existing, static, post-hoc green budgeting systems restrict successful management of the public money. To fill this vacuum, this study introduces an AI-based Green Budget Optimizer to sustainable financial planning with the focus on the Air Quality Index (AQI) as a dynamic proxy of the environmental policy effectiveness. This paper fills a severe absence of unified models of computation by creating an AQI forecasting system using machine learning algorithms that can synthesize local green fiscal interventions in five vastly polluted Indian megacities. Training with early historical data showed that there was serious data leakage and gross cross-validation instability. As a result, the paradigm changed to be a multi-factor causal structure to model AQI by strictly using engineered emission, industrial, vehicular, waste, and policy factors. The intended framework will reach the high-confidence level of the generalization threshold ($R^2 \approx 0.9$) by mathematically encoding socio-economic policies into dynamic optimization levers. This predictive engine, when realized, will objectively measure the AQI increases with each funding unit and green budgeting will cease to be retrospective accounting and instead a mathematically rigorous science in order to optimally utilize municipal capital.

Index Terms—Climate Mitigation, Public Financial Management (PFM), Machine Learning, Air Quality Index (AQI), Green Budgeting

I. INTRODUCTION

A paradox of the 21 st century is that the world is experiencing a high rate of technological innovation, and at the same time, the global ecological systems are destabilizing [3]. With the government undergoing the ripple effects of climate change and urban pollution, the need to adopt green finance has taken a critical turn [10]. Nonetheless, there is a systemic gap: the requirement of climate financing is significantly higher than the access to public funds [9]. Green budgeting should be effective by moving away a retrospective accounting practice to a mathematically optimized method of capital deployment [1].

Existing green budgeting models are also based more on subjective classifications and past experience instead of actual, forecasting impact evaluation [4]. Although Artificial Intelligence (AI) also introduces the ability to model the effects of policies with the help of so-called digital twins, the combination of machine learning, localized environmental policy, and efficient financial optimization has not been properly studied yet [12]. This paper will fill this gap with a proposal of an AI-based Air Quality Index (AQI) Prediction and Green

Policy Simulation Model, where AQI is a measurable proxy of environmental policy effectiveness [8].

Air pollution in cities is a global health crisis, which has been estimated to claim the lives of about 4.5 million people every year [3]. In India, the National Clean Air Programme (NCAP) targets the reduction of the particulate matter in 40% by 2026 [10]. Although the infrastructure surveillance has grown (the number of PM2.5 stations has grown by 344% since 2017), data integrity still poses a challenge as a result of maintenance problems and sensor drift [5]. Policy attribution is the biggest obstacle that municipal budgeting faces [1]. Historically, it is hard to distinguish the effect of particular spending (e.g., NCAP funds) due to other national policies such as transition to BS-VI vehicular standards or the increase in LPG availability [9]. The existing models charge expenditures to be incurred once they are approved, and they do not estimate where the marginal dollar will create the greatest reductions in pollution [4].

The optimizer proposed combines three emerging disciplines:

- Green AI: Beyond energy-efficient computing to AI that creates macro-environmental behavior [2].
- Performance-Based Budgeting (PBB): AI is used to maximize future returns in an uncertain environment [1].
- Robust Optimization: The use of mathematical models such as the BertsimasSim formulation to make the fiscal feasibility in the case where environmental variables vary dramatically [12].

The current literature tends to assume that the environment is a fixed financial asset category [11]. It is considered in this study as an oscillating reacting physical system that fills the disjunction between computational efficiency and policy simulation in the real world [3]. The first stage of the research focused on a generalized green budget model in various fields (water, soil, air) [7]. Nevertheless, disaggregated municipal data implied that a priority should be given to AQI-based modeling [8]. AQI is a standardized, high-frequency data stream, which is a direct measure of both the health of the population and policy effectiveness [10].

The initial modeling experiments involving historical autoregressive AQI data (predicting tomorrow based on today) were unsuccessful because of model leakage and extreme cross-validation instability (R^2 scores between -2.5 and 0.8) [6]. To address this, the approach shifted to a multi-factor

causal model, and AQI was modeled on five engineered types [2]:

Emissions: GHG emissions by industries and commerce [9].2. Industrial Development: Measures of manufacturing development [4].3. Vehicle Density: Proxy of NOx and PM emissions by internal combustion engines [5].4. Waste Metrics: Signs of localized burning of solid municipal waste [13].5. Policy Indicators: Dynamics that demonstrate green actions (e.g., EV subsidies, emission caps) [1].

- Industrial Growth: Metrics of manufacturing expansion [4].
- Vehicle Density: Proxies for NOx and PM emissions from internal combustion engines [5].
- Waste Metrics: Indicators of localized municipal solid waste burning [13].
- Policy Indicators: Dynamic variables representing green interventions (e.g., EV subsidies, emission caps) [1].

The model was able to stabilize to a leakage-free baseline by synthetic generation on 50,000 samples with a 15-fold cross-validation of 5,000, constituting an increase of 10x over the baseline of 5,000 samples [6]. The present goal is to use Bayesian hyperparameter optimization to achieve high confidence (R^2) of 0.9 to a high level of confidence [12]. The last framework acts as a forecasting engine that is sensitive to policy [1].1. The socio-economic variables play the role of mathematical optimization levers [11].2. Predictive Layer is a metric that measures the anticipated improvement of AQI per unit of funding [8]. In this system:

- Socio-economic variables act as mathematical optimization levers [11].
- The Predictive Layer quantifies the expected AQI improvement per funding unit [8].
- The Optimization Layer proposes certain budget tranches that will give an optimal ROI to the environment [2].

The fast-growing climate crisis requires an exit of fiscal guesswork [3]. The AI-Based Green Budget Optimizer uses the policy variables as optimization levers to make green budgeting an exact science [4]. It enables Urban Local Bodies (ULBs) to develop bankable projects to attract the use of the private capital through municipal green bonds by ensuring unquestionable evidence is presented about the impact on the environment [9].

In the end, such a structure will make sure that all units of both public and private capital are put in place with mathematical accuracy to guarantee an ecologically stable and sustainable future [5].

II. LITERATURE SURVEY

One of the significant contributions of this paper is the combination of the indicators of emissions and pollutant data giving a platform to policy based environmental modelling. But due to the lack of data and the inadequate conditions of emission data coverage, the model could not learn long-term correlations and this fact influenced the predictability results. Thus, the study may not be viewed as a definitive

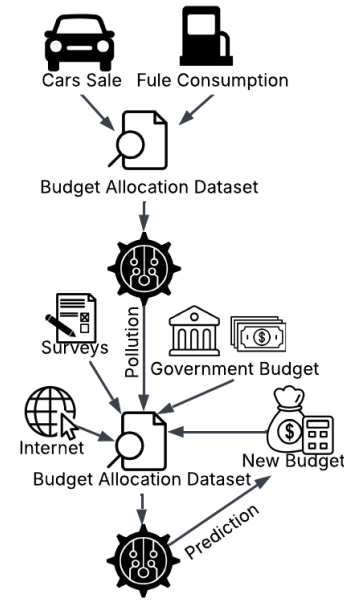


Fig. 1. visualization of working of proposed framework

predictive algorithm, but rather as a methodology framework. Having bigger multi-year datasets, detailed emission inventories, and meteorological variables, the future work will result in better prediction accuracy and allow planning green budget and developing environmental policies much more productively with the help of AI.

The available literature on the topic of sustainable artificial intelligence and decision systems based on optimization can be broadly categorized into two major directions: Green AI, which is concerned with minimizing the computational footprint, and frameworks based on AI to improve budget and resource allocation [3]. Even though both aspects have had major improvements, they are not collaborating with each other and fail to solve the issue of environmental intelligence and financial decision-making [1].

First one is oriented toward Green AI, in which the primary aim is to reduce the power use and the carbon footprint of the machine learning models [14]. The multi-objective optimization methods have been suggested in a number of studies, and they presuppose that there is a trade-off between model accuracy and computational efficiency and environmental impact [2]. They are frequently the carbon emissions as a parameter when training a model, and accent of lifecycle-on energy evaluation [15]. Although such techniques can be applied to make AI systems more sustainable, the techniques are largely geared towards computing processes optimisation and are not applicable to real-world applications in the environment like air quality prediction or policy evaluation [6].

In parallel to this, there is a second set of studies that researches the application of AI-based optimization of the budgeting and governance systems [1]. To enhance the decision making under uncertainty, in this region, hybrid models which combine a combination of machine learning, including

neural networks and strong optimization frameworks have been proposed [1]. The advantage of these systems is that they result in a better distribution of budgets since the financial variables are predicted and countered against fluctuations [4]. Also, simulation- and cloud-based methods have been developed to promote scalability and forecasting performance [8]. Nevertheless, these frameworks mainly focus on financial performance and risk management with the exceptionally minimal focus on environmental aspects such as the level of pollution or Air Quality Index (AQI) [9].

In both directions of the research, the optimization strategies like robust optimization and multi-objective decision making have been popular in addressing the issue of uncertainty and trade-offs [2]. The approaches enable the systems to remain viable in various states and balanced to a set of various factors like performance and efficiency [12]. Although these techniques are effective though, they are often applied on domain specific boundaries and are rarely combined with environmental prediction models [6].

The gap in the existing literature is significant in this review [3]. Existing methods of Green AI are concentrated on the sustainability of computation and do not regard the modeling of the environmental outcome, whereas the AI-based budgeting systems are not linked to the real-time environmental indicators [14]. Moreover, the absence of consistent frameworks that unite predictive environmental analytics, e.g. AQI estimation, with policy simulation and budget optimization [15] can be mentioned.

Thus, there is an evident necessity of a unified procedure that will relate these fields [10]. To address this gap, the proposed Green Budget Optimizer framework combines AQI prediction done with machine learning with policy simulation and greener budget allocation to provide the opportunity to make data-driven and environmentally-aware decisions [11].

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III. METHODS

The primary objective of this study is to develop a machine learning-based AQI prediction framework and extend it into a

TABLE I
LITERATURE REVIEW SUMMARY OF ML-BASED SUSTAINABLE BUDGET OPTIMIZATION APPROACHES

Paper & Date	Key Achievements	Limitations
Chukwuma-Eke et al. (2021)	Integrated ML with Monte Carlo simulations for cost/risk forecasting; incorporated green financing and carbon credits into budget management.	Primarily a conceptual framework; requires further empirical validation across diverse geographical contexts.
Gupta et al. (2023)	Developed a MOBO-based approach to balance accuracy/efficiency with holistic carbon footprints (operational and embodied).	Evaluating sustainability objectives remains time-consuming; experimentation costs can still be substantial.
Mohd Yaacob (2024)	Achieved 92.4% accuracy in risk/budget optimization using GBMs; utilized GreenCloud to reduce operational carbon footprints.	High complexity in integrating with legacy infrastructure; requires highly skilled personnel for system maintenance.
Acharya et al. (2025)	Optimized model training schedules to fit within predefined sustainability budgets; focused on environmental footprint reduction.	Limited focus on post-deployment inference sustainability; primarily focused on the training phase.
Ranpara (2025)	Demonstrated 25% energy reduction and 18% resource recovery improvement; integrated Blockchain for transparent material tracking.	High initial computational overhead for system setup; performance is dependent on high-quality real-time IoT data.
Khadivar & Saba (2026)	Improved budget deviation indicators through hybrid ANN-Robust Optimization; focused on performance-driven rather than cash-driven allocation.	The "black-box" nature of ANNs can conflict with governance transparency; requires the addition of Explainable AI (XAI) for accountability.

policy-sensitive forecasting system that enables simulation of green policy impacts. The framework is designed to accurately predict the Air Quality Index (AQI), incorporate policy-related variables influencing air pollution, simulate the effects of green interventions, and provide a foundation for green budget optimization. The study initially explored general green budget optimization, but due to insufficient and inconsistent budget allocation data, the problem was reformulated into a more structured approach focused on AQI-centered modeling. AQI was selected as the measurable environmental outcome variable through which policy effectiveness could be quantified.

AQI datasets were collected for five Indian cities:

- Bangalore
- Delhi
- Hyderabad
- Kolkata
- Mumbai

The initial dataset contained approximately 5,000 observations.

```

Combined DataFrame shape: (12042, 22)
  City  Date  AQI  PM2.5  PM10  NO2  SO2  CO  O3  \
0  Bangalore  01/01/18  68.0  37.4  73.44  56.44  77.52  0.71  64.6
1  Bangalore  02/01/18  76.0  41.8  82.08  63.08  86.64  0.80  72.2
2  Bangalore  03/01/18  70.0  38.5  75.60  58.10  79.80  0.74  66.5
3  Bangalore  04/01/18  78.0  42.9  84.24  64.74  88.92  0.82  74.1
4  Bangalore  05/01/18  82.0  45.1  88.56  68.06  93.48  0.86  77.9

  Unnamed: 9  ...  Unnamed: 12  Unnamed: 13  Unnamed: 14  Unnamed: 15  \
0  NaN  ...  NaN  NaN  NaN  NaN  NaN
1  NaN  ...  NaN  NaN  NaN  NaN  NaN
2  NaN  ...  NaN  NaN  NaN  NaN  NaN
3  NaN  ...  NaN  NaN  NaN  NaN  NaN
4  NaN  ...  NaN  NaN  NaN  NaN  NaN

  Unnamed: 16  Unnamed: 17  Unnamed: 18  Unnamed: 19  Unnamed: 20  \
0  NaN  NaN  NaN  NaN  NaN  NaN
1  NaN  NaN  NaN  NaN  NaN  NaN
2  NaN  NaN  NaN  NaN  NaN  NaN
3  NaN  NaN  NaN  NaN  NaN  NaN
4  NaN  NaN  NaN  NaN  NaN  NaN

  Unnamed: 21
0  NaN
1  NaN
2  NaN
3  NaN
4  NaN
[5 rows x 22 columns]

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Fig. 2. visualization of working of proposed framework

Additionally, policy-related factors and environmental drivers influencing AQI were identified, including:

- Carbon emissions
- Vehicle sales

- Industrial development
- Waste production
- Government AQI improvement policies

These variables were structured into a simplified and unified dataset. A machine learning model was then trained using this

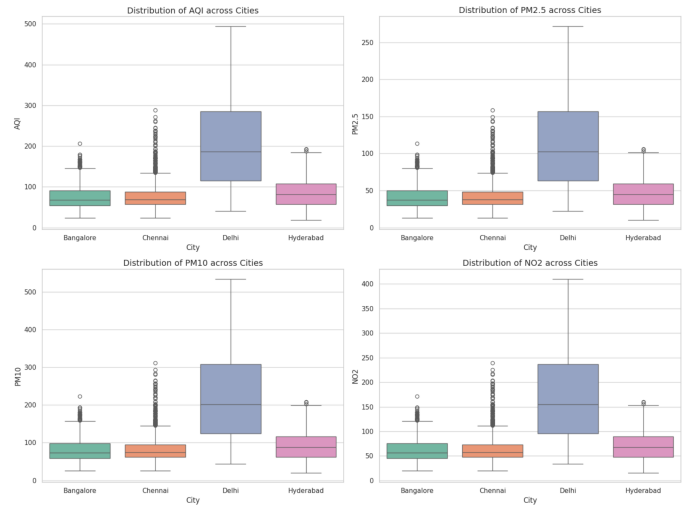


Fig. 3. visualization of working of proposed framework

engineered dataset and evaluated using 15-fold K-Fold cross-validation with R^2 score as the evaluation metric. The initial models that included policy variables produced an average R^2 score of approximately 0.4. The methodological objective is to achieve a stable R^2 score near 0.9 before integrating the predictive model into the optimization framework.

However, cross-validation results ranged from -2.5 to 0.8, with unstable averages, indicating insufficient generalization. The feature engineering is Based on poor predictive stability, the research methodology shifted from direct AQI modeling to a multi-factor causal structure.

AQI was modeled as dependencies shifts to:

- Emission-related factors

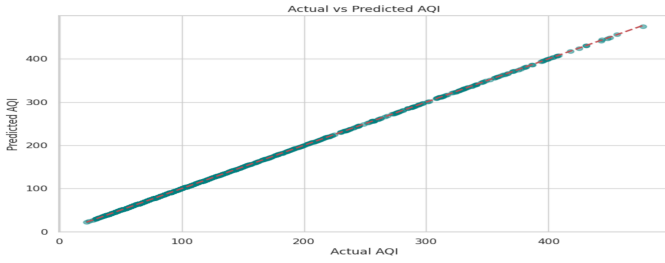


Fig. 4. visualization of working of proposed framework

- Industrial growth indicators
- Vehicle density proxies
- Waste metrics
- Policy indicators

A simplified structured dataset was constructed containing these engineered variables to better reflect AQI determinants.

The machine learning model was trained using the engineered dataset.

- Cross-validation: 15-fold K-Fold
- Evaluation metric: R^2 score

Initial models with policy inclusion produced an average $R^2 \approx 0.4$.

The methodological goal is to achieve a stable R^2 score near 0.9 before integrating the predictive engine into the optimization framework

Policy-Sensitive AQI Forecasting Framework proposed framework consists of:

1) **Predictive Layer**

A trained ML model that predicts AQI based on emission, industrial, vehicle, waste, and policy-related inputs.

2) **Policy Simulation Layer**

Policy interventions are encoded as structured feature adjustments (e.g., reduction in emissions, industrial regulation changes).

3) **Scenario Evaluation**

Modified inputs are passed through the trained model to estimate resulting AQI.

4) **Decision Output**

The predicted AQI improvements under different simulated policies provide a quantitative basis for green budget allocation

Final goal of the research is once the predictive model achieves strong generalization (target $R^2 \approx 0.9$):

- The model will serve as a forecasting engine.
- Policy variables will act as optimization levers.
- The system will quantify AQI improvements resulting from varying policy intensities.
- This will enable structured green budget optimization decisions.

IV. RESULT AND ANALYSIS

A. Dataset Preparation Outcomes

The dataset preparation outcomes consolidated as the dataset was created by combining Air Quality Index (AQI)

data from five major Indian cities: Bangalore, Chennai, Delhi, Hyderabad, and Mumbai. The data was preprocessed through date conversion, removal of redundant columns, and handling missing pollutant values to ensure consistency. To include broader environmental factors, a carbon emission dataset containing CO and methane emissions was added. Since the original dataset was unavailable, a placeholder dataset for 2018 was used. The national-level emission data was distributed across the five cities and converted into daily estimates, then merged with the AQI dataset by city and date. Missing values created during merging were handled using forward filling and removal of remaining incomplete records. Feature engineering was performed to improve prediction performance. The final dataset included lag features, rolling statistics, cyclic temporal features, and AQI volatility measures. The target variable was defined as AQI values 24 days ahead. After preprocessing, the final dataset contained 9,641 observations and 52 features, including emission-related variables such as Daily_CO2_Emissions_kt and Daily_Methane_Emissions_kt.

When we used Random Forest Model the Performance of the model was evaluated using RandomForestRegressor with time-series cross-validation. Initial results using a random train-test split showed very high accuracy, but this was due to data leakage. To correct this, TimeSeriesSplit validation was used, which resulted in lower but more realistic performance. After applying Recursive Feature Elimination and Grid-SearchCV, the optimized Random Forest model was evaluated using 50 chronological folds.

The resulting performance metrics were:

TABLE II
STATISTICAL SUMMARY OF R^2 SCORES

Metric	Value
Average R^2 Score	-0.1398
Standard Deviation	0.5438
Minimum R^2	-2.4931
Maximum R^2	0.7406

The negative mean R^2 score indicates that the model often performed worse than a baseline predictor, and the high standard deviation shows instability across time periods.

B. Temporal Variability in Model Performance

The R^2 scores varied significantly across the 50 folds. The model performed well in some time periods but poorly in others, indicating that AQI patterns change over time due to seasonal effects, weather conditions, and sudden pollution events. This shows that AQI data is non-stationary and difficult to predict over long horizons.

C. Feature Importance Analysis

Feature importance analysis showed that lagged pollutant values were the most important predictors, especially:

- AQI_lag_6
- PM2.5_lag_6
- PM10_lag_6

Other important features included rolling statistics, seasonal (cyclic) features, and AQI volatility indicators. These results align with environmental studies showing that recent pollution levels and seasonal patterns strongly influence AQI.

D. Impact of Emission Features

CO and methane emission features were successfully integrated into the model, but their impact on prediction performance was limited. This is likely due to the emission dataset covering only one year. Future studies using longer-term emission data may improve results.

E. Key Observations

The experimental results reveal several important insights regarding AQI forecasting:

- 1) Predicting AQI 24 days ahead is challenging.
- 2) Recent pollutant levels are the strongest predictors of AQI.
- 3) Seasonal patterns significantly affect AQI.
- 4) Model performance varies across time, showing non-stationary behavior in air pollution data.

CONCLUSION

Blockchain is a game-changer in academic credentialing as a safe, open, and decentralized alternative to the usual process of issuing, storing, and authenticating academic credentials. With digital credentials increasingly depending on technology, it has served to get people to see just how useless the existing systems are with their forgery of credentials, lag in verification, theft of data, and students not having ownership of their accomplishments. A blockchain platform resolves such problems by making academic records tamper-proof, simple to verify, and available anywhere in the world without an intermediary. Utilizing smart contracts, cryptographic hashing, and decentralized storage, blockchain technology allows institutions to issue tamper-evident and unalterable verifiable credentials. Students gain more control of their academic credentials since they are able to save and distribute them safely to employers and institutions with blockchain-verified QR codes, digital wallets, or DIDs. The transition avoids the need for third-party verification agencies, thereby saving employers and universities time and money.

Direct deployment of blockchain credentialing has also started. Blockcerts by MIT, India's DigiLocker, European Blockchain Partnership, and Blockchain for Education from IBM are evidence of the practicality of decentralized control systems of certificates[6]. These proofs ensure that blockchain is able to enable real-time verification, enhanced security, and global interoperability—ingredients core to the process of modernization of academic transcripts. While these have been improved, there are still issues. Scalability problems, privacy, regulatory problems, and institutional adoption need to be solved prior to blockchain becoming the norm for credential validation. With advancements in blockchain technology, further research, regulatory innovation, and cross-industry partnerships will be at the forefront of determining

the future of blockchain credentialing. Effective deployment of scalable, privacy-centric, and law-abiding blockchain credentialing systems will revolutionize documentation, authentication, and certification of academic credentials in the digital era. As decentralised identity tech, interoperability protocols, and privacy-preserving technologies advance, academic qualifications issued on blockchain can yet prove themselves as the blocks to construct an international education system[6] instilling faith, efficiency, and transparency in generations to come.

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